

STEP-BY-STEP PANEL CHECK IN-STORE

When you get a chance to inspect a laptop in person, here are some quick checks:

- 1. Dead pixels / uniformity:** Open or ask to open a plain background (you can use Paint or a browser fullscreen). Check a solid white screen and a solid black screen. On white, look for any tiny black or off-color dots (dead or stuck pixels). On black, in a dimmer area if possible, see if there are any bright pixels. A few stuck pixels can happen; most policies won't replace for fewer than 3–5, but it's good to know before you buy. Also notice if the black screen is truly black or if light leaks at the edges.
- 2. Backlight bleed (for IPS/LED screens):** In a dark room (or cup your hands around the screen) display a pure black image at medium brightness. Check if the corners or edges have a "glow." A slight glow is common on IPS (so-called IPS glow), but large, uneven patches of light are backlight bleed. Minor bleed is usually not an issue in daily use, but if you see big bright blooms on black, that panel might be sub-par.
- 3. Brightness and colours:** Turn the brightness up and down. At max, does it look bright enough for you? (Many stores are brightly lit – if it looks dull in store, it will be dull at home). Observe color punchiness – open a high-resolution photo or a YouTube nature video in 4K if possible. This will give you a sense of gamut and contrast. On an OLED, you'll notice very deep blacks and vivid colors. On a good IPS, colors should still look rich (not washed out). If comparing two models side by side, note which has more "pop," but also which looks more natural (some screens oversaturate by default).
- 4. Viewing angles:** Stand off-center. A quality IPS or OLED should remain fairly accurate in color and brightness even at a sharp angle. If you see a drastic shift (like it fades or inverts colors), it might be a

cheaper panel (or a TN panel, which you should generally avoid for a laptop in 2025). Some premium business models have privacy filters (e.g. HP SureView) that intentionally narrow the viewing angles – great for privacy, but those can make the screen look dimmer or with a slight haze front-on. Decide if that trade-off is worth it.

- 5. Check for PWM/flicker:** This one's tricky in store, but you can do the smartphone camera test. Lower the brightness to 50% or so, point your phone camera at the screen (slow-mo mode can help) – see if you notice flickering lines/bands on the phone screen. If yes, the screen likely uses PWM. High-frequency flicker might not show clearly on phone cameras, so this isn't foolproof, but strong banding indicates low-frequency PWM (could be an issue if you have sensitive eyes).
- 6. Touch and gloss vs matte:** If it's a touchscreen, test responsiveness and also see how reflective the glass is. Glossy touchscreens (common on 2-in-1s and many OLEDs) give punchier image quality but reflect lights like a mirror. Matte screens cut reflections – if you're often in bright offices or cafes, a matte IPS might actually be more comfortable than a glossy OLED despite the latter's superior contrast. Decide based on your typical environment.

By doing these checks, you'll ensure the display meets your quality standards. Remember, the display is one of the most important components – a high-resolution, colour-accurate, non-flickering screen can greatly enhance your computing experience (and a poor screen will bother you every single day). It's worth prioritizing: e.g., choose a 14-inch model with a gorgeous 2.8K OLED 90Hz panel over another with a mediocre 1080p LCD, even if the latter has a slightly faster CPU, unless your work is purely CPU-bound. You look at the screen every moment, so don't skimp on it.

tions, and of course gaming. Even for general use, a 90Hz or 120Hz display makes the system feel more responsive. If you're a gamer, look for at least 120Hz. Many gaming laptops have 144Hz or 240Hz screens (1080p or 1440p). Some creative laptops also offer 120Hz OLEDs (making digital pen input smoother, etc.). Keep in mind driving high refresh can use a bit more power. Some laptops let you switch between 60Hz and the higher rate to save battery when needed. If you see a spec like "360Hz FHD panel" – that's purely for competitive gaming (esports titles). For most, 120–165Hz is plenty smooth.

If display quality (color, resolution) is more important to you than ultra-high refresh, it's okay to get a 60Hz – for example, a 4K 60Hz OLED for a photographer rather than a 1080p 144Hz TN panel (which sacrifices resolution and color just for speed). Choose based on your primary use.

- **Brightness:** Measured in nits (cd/m²). A typical laptop is ~300 nits. For indoor use, 250–300 is fine. If you work in brighter environments or near windows, 400+ nits helps. Premium laptops often have 500-nit screens. HDR-capable models might boast 600–1000 nits peak (useful for HDR